

DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078

Department of Artificial Intelligence and Machine Learning



INTERNSHIP REPORT

ON

“Next-Life: AI Health App”

Submitted in partial fulfillment for the award of degree(21INT82)

BACHELOR OF ENGINEERING IN

Department of Artificial Intelligence and Machine Learning

Submitted by:

Jyotishmati (1DS21AI025)

Conducted at



Studio Ilios(Design & Engineering)

Department of Artificial Intelligence and Machine Learning

Dayananda Sagar College of Engineering Bangalore-78

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CERTIFICATE

This is to certify that the Internship titled “**Next-Life: AI Health App**” carried out by **Jyotishmati [1DS21AI025]**, a bonafide student of Dayananda Sagar College of Engineering, in partial fulfillment for the award of **Bachelor of Engineering**, in **Artificial Intelligence and Machine Learning** during the year 2024-2025. It is certified that all corrections/suggestions indicated have been incorporated in the report.

The internship report has been approved as it satisfies the academic requirements in respect course Internship (21INT82)

**Signature of Reporting
Head**

Mr. Arjun

CEO of the
Studio Ilios

**Signature of Internship
Coordinator**

Prof. Anupama
Dept. of AIML
DSCE

Signature of HoD

Dr. Vindhya P Malagi
Dept. of AI & ML
DSCE

External Viva:

Name of the Examiner

1) _____

2) _____

Signature with Date

DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078

Department of Artificial Intelligence and Machine Learning



DECLARATION

I, **Jyotishmati**, final year student of Artificial Intelligence and Machine Learning, Dayananda Sagar College of Engineering, Bangalore - 560 078, declare that the Internship has been successfully completed, in. This report is submitted in partial fulfillment of the requirements for the award of Bachelor's degree in Artificial Intelligence and Machine Learning, during the academic year 2024- 2025.

Date:

USN: 1DS21AI025

Place: Bangalore

NAME: Jyotishmati

OFFER LETTER

Studio - ILIOS

DESIGN AND ENGINEERING

Internship Offer letter Confirmation

Date 28 2 2025

Dear **Jyotishmati**, 4th year, 8th semester at, **Dayananda Sagar University, Bangalore**.

We are pleased to offer you an internship position at **Studio ILIOS** as a **Software Development Intern**. This internship will be based in **Bangalore** and will run for a duration of **4 to 5 months**, starting from **February 10 to May or June**, with the potential for extension based on performance and project requirements.

Role and Responsibilities:

During your internship, you will be working closely with our development team and will gain hands-on experience across multiple technologies and domains, including:

Frontend Development: Building dynamic and responsive user interfaces using **React.js** and **React Native** for web and mobile applications.

Backend Development: Developing robust backend systems, REST APIs, and microservices using **Node.js**, **Express.js**, and integrating with **MongoDB** for data storage and management.

AI & ML Model Integration: Assisting in the development, training, and deployment of **AI/ML models**, and integrating them into web and mobile applications.

Blockchain Technology: Exploring and integrating **decentralized applications (dApps)** using blockchain technologies, including **Ethereum**, **Polygon**, **Solana**, or other relevant platforms.

Collaborative Development: Working with a cross-functional team, participating in daily stand-ups, contributing to code reviews, and sharing innovative ideas.

Version Control: Using **Git** and collaborative tools like **GitHub** or **GitLab** for version control and collaborative development.

Compensation & Benefits

We are pleased to offer you a internship stipend every month.

Confidentiality:

As an intern, you will have access to proprietary information and intellectual property. You are required to maintain strict confidentiality during and after your internship and sign a Non-Disclosure Agreement (NDA) if required.

Upon successful completion of the internship, you will receive an **Internship Completion Certificate** from **Studio ILIOS**, highlighting your contributions and technologies worked on.



ARJUN Founder, CEO
Studio-ILIOS, Bangalore

102 / 4, 3rd floor Opposite to Royale Concorde School, Chelikere Kaylan Nagar Shankar,
Bengaluru, Karnataka 560043 – E-mail: design@studioilios.com Ph- 9916550088

COMPLETION CERTIFICATE

ACKNOWLEDGEMENT

This Internship is a result of accumulated guidance, direction and support of several important persons. We take this opportunity to express our gratitude to all who have helped us to complete the Internship.

We express our sincere thanks to our Principal **Dr. B G Prasad**, for providing us adequate facilities to undertake this Internship.

We would like to thank **Dr. Vindhya P Malagi**, Head of Department – AI & ML, for providing us an opportunity to carry out Internship and for her valuable guidance and support.

We would like to thank **Mr. Arjun**, for guiding me during the period of internship.

We express our deep and profound gratitude to our Internship Guide, **Prof. Anupama**, Assistant Professor – AI & ML, for her keen interest and encouragement at every step in completing the Internship.

We would like to thank all the faculty members of AI & ML department for the support extended during the course of Internship.

We would like to thank the non-teaching members of the Department of AI & ML, for helping us during the Internship.

Last but not the least, we would like to thank our parents and friends without whose constant help, the completion of Internship would have not been possible.

Jyotishmati [1DS21AI025]

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CHAPTER 1

INTRODUCTION

1.1 Introduction to Next Life

Next Life is an AI-powered mobile health application built to revolutionize how individuals manage and interact with their healthcare. In a world where medical systems are often overwhelmed and access to timely care can be challenging, Next Life aims to empower users with intelligent, personalized health support—right at their fingertips.

The app integrates advanced machine learning models trained on authentic medical datasets with a clean, intuitive design to provide accurate, real-time assistance. Whether users are tracking symptoms, analyzing reports, or seeking quick answers to health concerns, Next Life delivers AI-generated insights that can often be more data-driven and consistent than traditional approaches.

A standout feature of the app is the customizable multi-agent assistant, which allows users to tailor their digital health companion based on their specific needs—whether it's mental health, fitness, chronic condition management, or everyday health tracking.

Why Next Life

The core idea behind Next Life is to make healthcare smarter, more accessible, and less dependent on physical visits. Millions of people visit hospitals for basic concerns or checkups that could be handled digitally. Long queues, delayed diagnostics, and limited access to specialists can often hinder timely intervention.

- **Reducing unnecessary hospital visits:** Through intelligent AI-powered consultations and health tracking.
- **Providing more accurate health insights:** by analyzing medical reports, scans, and health data using ML models—often identifying patterns that might be overlooked in manual diagnosis.
- **Offering a handy, on-the-go solution:** that makes it easier for users to take control of their health anytime, anywhere.

With this app, we aim to **augment—not replace—traditional healthcare** by serving as a first layer of support, ensuring users get preliminary guidance, predictions, and alerts before things escalate.

Core Objectives

Accessibility: Bring healthcare tools to the user—anytime, anywhere.

Personalization: Multi-agent assistants tailored to each individual's lifestyle and health concerns.

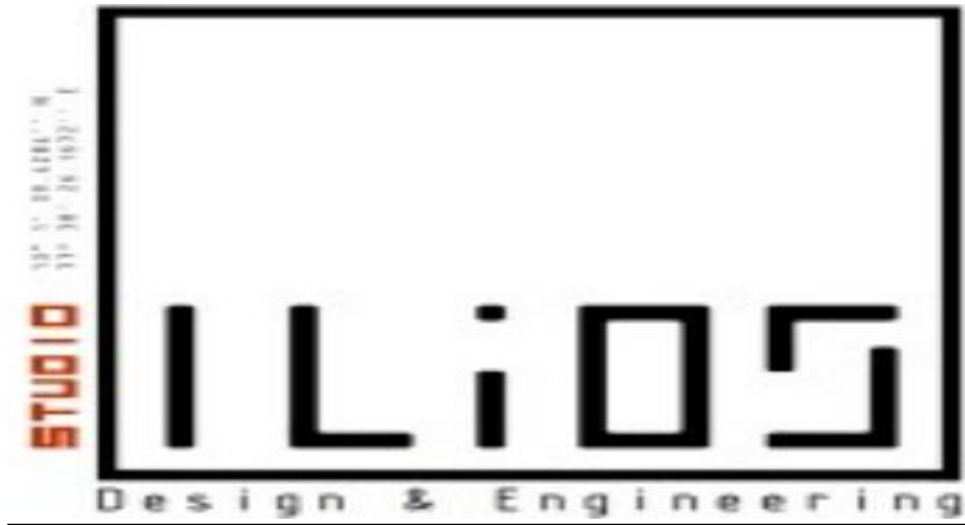
Accuracy: Use trained machine learning models to analyze user data and provide intelligent responses.

Convenience: A mobile-first approach that makes managing one's health effortless and efficient.

CHAPTER 2

DESCRIPTION OF ORGANIZATION

Studio Ilios - Design & Engineering



2.1 Introduction

Studio Ilios is a multidisciplinary design studio based in Bengaluru, Karnataka, India. Founded in 2017, the studio specializes in architecture design, interior design, automotive design, and landscape design. With a team of 2–10 employees, Studio Ilios operates as a partnership.

The studio is actively involved in integrating technology with design to create innovative solutions. Their recent projects focus on AI-powered cities, sustainable mobility, and next-generation urban design. Studio Ilios emphasizes creating spaces that are resilient, adaptable, and environmentally friendly.

In addition to their design projects, Studio Ilios is developing a health application aimed at improving healthcare services through technology. The app focuses on enhancing user experience and accessibility in the healthcare industry. The mission is to merge technology and healthcare to provide effective and seamless solutions, with a strong emphasis on user-centered design.

For this initiative, Studio Ilios is offering a Software Developer Intern position. The role involves designing a health application using current web and mobile technologies. Responsibilities include creating responsive user interfaces with React.js and React Native, developing robust REST APIs and microservices using Node.js and Express.js, and

managing data storage with MongoDB. Version control is maintained through Git, utilizing platforms like GitHub or GitLab for collaborative development.

Services Offered:

Studio Ilios is a multidisciplinary design studio based in Bengaluru, India.

- **Architecture Design:** Crafting innovative and sustainable architectural solutions.
- **Interior Design:** Designing functional and aesthetically pleasing interior spaces.
- **Automotive Design:** Exploring futuristic automotive concepts and designs.
- **Landscape Design:** Creating harmonious outdoor environments that complement architectural projects.

Growth:



Career Opportunities and Culture:

Studio Ilios fosters a high-performance culture emphasizing customer delight and integrity. The company offers a stimulating work environment with provisions for self-learning and growth, promoting transparency, collaboration, and opportunity.

Mission: To empower organizations with secure, scalable, and sustainable living.

Vision: To revolutionize everyday living by seamlessly integrating design, technology, and artificial intelligence—empowering individuals through intelligent, accessible, and sustainable solutions in healthcare and beyond.

2.2 Founders

Architect AJ – CEO

Arjun Nagesh is the Founder and CEO of Studio Ilios, a multidisciplinary design studio

based in Bengaluru, India. He has a background in architecture, having studied at the University School of Design in Mysore, affiliated with Visvesvaraya Technological University.

Arjun's interests span various domains, including automotive design, architecture design, energy innovation, clean energy, product design, future cities, and future mobility. He has shared his design explorations and research and development in the automotive world, emphasizing the integration of sketches and AI-generated ideas.

In addition to his architectural pursuits, Arjun is also an entrepreneur with interests in filmmaking. He is currently involved in a project named "TimeMetal," which is in the building process.

CHAPTER 3

SCOPE OF WORK

The scope of work for Next Life: AI Health App encompasses full-stack development and integration of machine learning capabilities to deliver a smart, accessible, and predictive healthcare experience. The app reduces dependency on physical hospital visits by providing intelligent predictions based on medical data such as reports and scans. It is designed with scalable infrastructure and a user-first approach using modern frontend, backend, and cloud technologies.

1. Frontend Development - Cross-Platform Mobile Interface

- **Objective:**
Design and implement a user-friendly, responsive, and intuitive health app interface using React Native and TypeScript.
- **Methodology:**
Used React Native + Expo for cross-platform compatibility.
Modular components for reuse across screens.
State management and navigation using React Navigation.
Styled using native styling and TailwindCSS utilities.
- **Results:**
Delivered a seamless mobile experience, Improved usability and accessibility for healthcare services.

2. Backend Logic Implementation

- **Objective:**
Enable seamless communication between frontend and backend with secure and scalable APIs.
- **Scope:**
Design RESTful APIs for authentication, medical data uploads, user profiles, and ML integration, Implement middleware for file validation and security, Manage backend logic for report prediction and response generation.
- **Methodology:**
Developed using Node.js + Express.js
Modular API design with routes for multi-agent control, medical file parsing, and

user queries

Used JWT for secure authentication and role-based access control

Integrated response handlers for ML prediction results

- **Results:**

Real-time user-agent interaction enabled

Ensured secure medical data handling

Simplified communication between app and model endpoints

3. Machine Learning Integration

- **Objective:**

Use medical data to provide predictive analytics and intelligent responses for personalized healthcare insights.

- **Scope:**

Integrate trained ML models for medical report analysis

Generate user-specific predictions with higher precision than traditional diagnostics

Customize agent responses based on user condition

- **Methodology:**

Models trained on domain-specific datasets (medical scans, reports)

Integrated via cloud-hosted REST endpoints

JSON-based structured inputs/outputs with confidence scores and prediction labels

- **Results:**

Deliver early diagnostics and actionable insights

Reduced need for routine hospital visits

Supported doctors and users with secondary opinion mechanisms

4. MongoDB Integration – Secure Data Storage

- **Objective:**

Create a persistent, scalable, and cloud-connected data layer to store user profiles, reports, and predictions.

- **Scope:**

Store user data, agent preferences, prediction history, and chat logs

Enable search and filtering of past reports

Secure sensitive health data

- **Methodology:**

Integrated MongoDB Atlas as a cloud NoSQL DB

Designed collections for modular access (users, uploads, predictions, agents)

Implemented CRUD operations with Mongoose

- **Results:**

Achieved scalable storage with efficient access

Supported real-time updates and personalization

Complied with secure data handling practices

CHAPTER 4

LEARNING OBJECTIVES

1. Full-Stack Development Proficiency

- Understand the end-to-end process of building a cross-platform mobile app using React Native, Expo, and TypeScript.
- Gain hands-on experience in designing, building, and integrating RESTful APIs using Node.js and Express.js.
- Learn to manage real-time communication between frontend and backend for smooth user interactions.

2. Database Management with MongoDB

- Develop skills in designing scalable database schemas tailored for healthcare applications.
- Learn how to use MongoDB Atlas for secure cloud-based storage, including data modeling and CRUD operations.
- Understand data handling best practices, especially in storing sensitive health records.

3. ML Model Integration & Predictive Analytics

- Learn how to integrate machine learning models trained on medical datasets into production-ready applications.
- Understand API-driven interaction between frontend and AI models for tasks like image/report analysis and predictions.
- Explore confidence scoring, structured output formats, and post-processing of ML predictions.

4. Customizable Multi-Agent Design

- Understand the architecture for building modular, AI-powered digital agents.
- Learn how to provide users with custom agent experiences by adjusting preferences and use-case focuses (e.g., diet, mental health, fitness).
- Gain insights into how agent personalization improves engagement and care accuracy.

5. Gain Exposure to Real-World Project Management

Lastly, the internship provided me with an opportunity to observe and participate in real-world project management workflows. I was included in planning discussions, feature reviews, and internal testing phases. I worked closely with mentors and team members to fix bugs, update UI components, and refine user interactions based on feedback. I

maintained proper folder structures, consistent naming conventions, and development notes to ensure code clarity and maintainability. Through this experience, I gained a deep appreciation for agile development practices and learned how even minor UI/UX changes can significantly improve product usability in professional environments.

CHAPTER 5

CHALLENGES AND SOLUTION

1. Responsiveness Across Devices

Challenges:

Building a health app that delivers a consistent and accessible experience across various screen sizes (phones, tablets) was a key challenge.

Solutions:

Used React Native with Expo for cross-platform compatibility (iOS & Android).

Implemented responsive layouts using Flexbox and utility-first styling (like Tailwind CSS equivalents).

Tested across multiple screen sizes and used platform-specific adjustments to maintain visual consistency.

2. ML Model Integration with Real-Time UI

Challenges:

Integrating machine learning models trained on medical data with real-time predictions and ensuring latency-free UI rendering was complex.

Solutions:

Hosted ML models on cloud servers with RESTful APIs for prediction handling.

Created async request handlers and loading indicators in the UI.

Used confidence scoring and structured outputs to make predictions user-friendly.

3. Secure Handling of Sensitive Medical Data

Challenges:

Storing and transmitting personal health records, reports, and scans demanded a strong focus on security and compliance.

Solutions:

Implemented secure file upload validations (PDF, image formats only).

Used JWT-based authentication, HTTPS APIs, and MongoDB Atlas with access control.

Sanitized all inputs and handled backend requests with schema validation (e.g., using Joi or Mongoose schemas).

4. File Upload and Format Compatibility

Challenges:

Users upload various types of files (PDFs, reports, scanned images), which can vary in size, structure, and readability.

Solutions:

Implemented frontend validations for file size and type.

Backend processing pipelines included OCR and preprocessing layers to clean and format inputs before prediction.

Provided feedback messages for unsupported files and guided users with examples.

5. Custom Multi-Agent System Design

Challenges:

Building a flexible multi-agent system that adapts to users' individual needs (e.g., fitness, nutrition, mental health) required dynamic control.

Solutions:

Created a modular agent architecture with customizable user preferences.

Enabled agent-type selection and routing logic via backend APIs.

Designed UI components to support easy agent switching and personalization.

6. Cross-Team Collaboration & Version Management

Challenges:

Coordinating frontend, backend, and ML team efforts while maintaining version control and smooth integration was demanding.

Solutions:

Used Git & GitHub for structured version control and PR-based workflows.

Maintained clear documentation and consistent API contracts.

Held regular standups to align progress across teams.

Handled asynchronous requests correctly in React and aligned the data payloads for seamless communication between the UI and the server.

HARDWARE AND SOFTWARE REQUIREMENTS

Software requirements:

Operating System: Windows 10/11

Development Tools:

- Visual Studio Code/Cursor AI
- Git
- Postman

Programming Languages and Frameworks:

Frontend:

- Language: Typescript
- Framework: React Native(used for UI development and component-based structure)
- Libraries Used:
 - Axios – for HTTP requests
 - React Toastify – for real-time error notifications
 - React Navigation – For app screen routing
 - Tailwind CSS/NativeWind– for quick UI styling and responsive design

Backend:

- Language: Typescript
- Framework: Node.js with Express.js
- API Architecture: REST APIs
- Authentication: JWT (JSON Web Tokens)

Database:

MongoDB Atlas – Cloud-hosted NoSQL database

Mongoose – ODM for MongoDB schema validation

Hardware requirements:

Processor Intel i5 / AMD Ryzen 5 Intel i7 / Apple M1/M2

RAM 8 GB 16 GB or more

Storage 256 GB SSD 512 GB SSD or higher

Graphics Integrated GPU Dedicated GPU (optional for ML)

**Mobile
Devices**

Android 8.0+ / iOS 13+

Android + iOS physical
devices

CHAPTER 6

ACHEIVEMENTS AND CONTRIBUTION

During my project work on the Next Life: AI Health App, I had the opportunity to contribute extensively to both frontend and backend development, working collaboratively with my team on building responsive mobile interfaces and robust server-side logic. I also explored and integrated open-source AI models like BERT for intelligent health prediction features. These contributions not only enhanced the app's performance, user experience, and functionality, but also significantly strengthened my full-stack development and machine learning integration skills.

1. Frontend Enhancements

One of my major responsibilities was improving the user interface to ensure a better experience for both new and returning users. I focused on making the portal more visually appealing, intuitive, and responsive.

- **UI/UX Improvement:** I redesigned certain components to ensure consistency in layout, color schemes, and element placement. The goal was to reduce visual clutter and simplify the user experience, especially during the issue submission process.
- **Image Format Validation:** To improve file input reliability, I added a validation mechanism to accept only specific image formats (JPEG, JPG, PNG). This reduced the risk of incompatible file types being uploaded, ensuring smooth image rendering and improving backend processing.
- **Real-time Feedback with Toast Messages:** To give users immediate visual cues on their actions (e.g., form submission, errors, successful uploads), I integrated toast messages using a notification library. This significantly enhanced interaction feedback, helping users stay informed about the status of their actions without navigating away from the page.

2. Backend Improvements

My backend contributions involved enhancing the logic behind issue creation and improving data handling to ensure accurate and user-customizable reporting.

- **Custom Severity and CVSS Scoring:** Previously, these fields were static. I added support for dynamic selection, enabling users to input and edit severity levels and

CVSS scores based on their assessment. This flexibility made the portal more aligned with real-world VAPT reporting standards.

- **Fixing Date Display Logic:** While uploading issues, the date field used to display incorrect timestamps. I debugged the issue and implemented logic to fetch and display accurate timestamps that align with the user's local time zone.
- **Auto-Save and Dynamic Issue Creation:** I implemented a dynamic form system that allows users to create multiple custom issues in real-time. To prevent data loss during network issues or accidental reloads, I also integrated an auto-save mechanism that temporarily stores data and repopulates it when the user returns.

3. Development, Testing, and Deployment Workflow

This phase of my work revolved around ensuring that the implemented features were stable, efficient, and well-integrated across the system.

- **Full-Stack Feature Development:** I built and tested key features combining **React** (frontend), **FastAPI** (backend), and **MongoDB** (database). This provided a hands-on opportunity to apply my learning in a real-world development setting.
- **Cross-Component Issue Validation:** I worked on verifying data flow and issue creation logic across frontend forms, backend APIs, and the database, ensuring smooth interaction and error-free submissions.
- **Code Optimization:** While reviewing the existing codebase, I identified unused or redundant fields/components in the UI and removed them to improve performance and reduce cognitive load for users.
- **Debugging & Git Workflow:** I regularly debugged features in the DEV environment, using browser tools, logs, and API traces. For collaboration and version control, I followed a Git-based workflow. I used **GitLab** to manage feature branches, conducted peer reviews, and staged changes before merging. Once validated, the final changes were pushed to **GitHub** for deployment.

Technology stack

The development of the VAPT Reporting Portal involved a modern and scalable technology stack that enabled full-stack web development, real-time testing, and smooth deployment processes.

Frontend Technologies

- **Typescript:** A popular JavaScript library used to build dynamic and responsive user interfaces for mobile devices. It enabled the development of reusable components and efficient rendering, improving the overall user experience.
- **TailwindCSS/Native-Wind:** These standard style-sheets were used to structure the application layout, implement logic, and apply styling to make the interface user-friendly and visually appealing.

Backend Technologies

- **FastAPI (Python):** A high-performance web framework used to build RESTful APIs. It enabled fast development with automatic data validation, asynchronous programming, and clear routing mechanisms to handle HTTP requests efficiently.

Database

- **MongoDB Atlas:** A cloud-hosted NoSQL database that stored user data, issue reports, and other dynamic content. It provided scalability, flexibility in schema design, and efficient storage for JSON-like documents.

API Testing

- **Postman:** Used to test backend API endpoints during development. It allowed sending custom HTTP requests, validating responses, and identifying bugs before integrating with the frontend.

Version Control

- **Git (GitHub & GitLab):** Version control was managed using Git. GitLab was used during the development phase for branching, staging, and merge requests, while GitHub was used for maintaining the final deployment-ready repository.

Notification System

- **React Toastify:** A lightweight and customizable library used to display real-time toast notifications to users. It was used to show success, error, and alert messages after form submissions or actions.

HTTP Requests

- **Axios:** A promise-based HTTP client used in React for sending asynchronous requests to the FastAPI backend. It enabled easy integration with API endpoints and efficient data handling between client and server.

Development Tools

- **VS Code:** The primary code editor used for writing and organizing code. It offered useful extensions for linting, formatting, and Git integration.

- **Terminal & Git Bash:** Used for command-line operations like running servers, pushing code, and managing version control through Git commands.

Development Workflow And Process

During my work on the Next Life: AI Health App project, I followed a structured and collaborative software development lifecycle to build and enhance features across both the frontend and backend. This process ensured the product remained stable, user-centric, and technically scalable while aligning with the project's healthcare-driven mission. Below are the key stages followed throughout the development process:

1. Problem Identification and Feature Planning

The development process began with identifying user needs, app feature gaps, or technical limitations through discussions with the team and reviews of the Figma prototype. Whether it involved adding new UI components, improving backend logic, or integrating AI models, all requirements were clearly documented. We focused on designing features that would reduce hospital visits, increase accessibility, and provide personalized health insights through AI.

2. Code Implementation – Frontend and Backend

a. Frontend Development:

Using React Native with TypeScript, I developed dynamic and responsive UI components optimized for both Android and iOS devices. I ensured accessibility and mobile usability, adhering to a consistent design system guided by the Figma mockups.

b. Backend Development:

On the server side, I used Node.js and Express.js to create RESTful APIs, handle user data, and integrate MongoDB Atlas for secure data storage. Backend code was modular, reusable, and designed to efficiently communicate with AI model endpoints.

3. API Testing Using Postman and Validation

To ensure the backend APIs functioned as expected, I tested all endpoints using Postman. This included validating input types, error messages, authentication mechanisms, and successful response formats. This step helped eliminate issues early before integration with the frontend.

4. Local and Cloud Environment Testing

Once tested independently, I performed integrated testing on my local development

environment using Expo Go for real-time preview. Later, we deployed test versions to cloud services for broader team feedback and device-specific testing. This helped catch responsiveness issues and ensured seamless behavior across various screen sizes and platforms.

5. Review and Approval Process

I used Git and GitHub for version control and collaboration. Each feature or fix was pushed to a separate branch and reviewed by teammates. Peer reviews focused on code quality, modularity, and adherence to project standards. Feedback was actively incorporated before merging.

6. Final Merge and Deployment

After all validations were complete, changes were merged into the main codebase. Deployment included syncing backend endpoints with the latest ML model outputs, ensuring the complete functionality from user input to AI-generated insights. The final app builds were prepared for internal testing before production deployment.

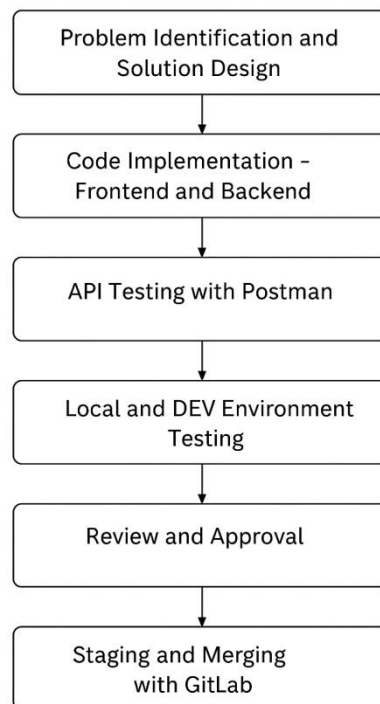


Fig 6.1: Development and Testing Process

Frontend:

1. Designed UI/UX to make it visually appealing and easy to understand for new users:

- Unnecessary UI components were removed to enhance clarity and performance.
- Themes and interactive features were added to create a more engaging user experience.

- Core UI elements were improved for better accessibility, feedback, and responsiveness.

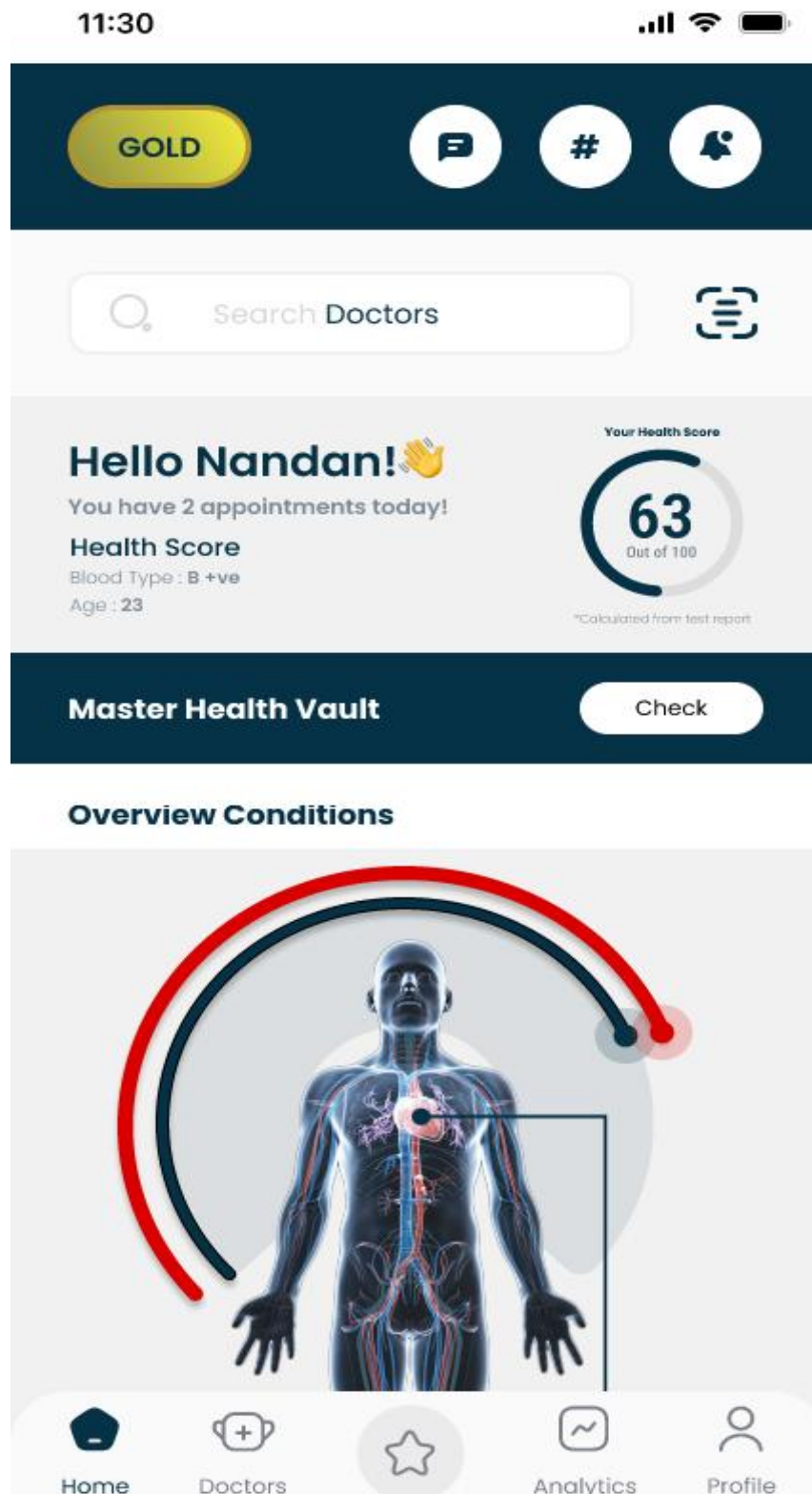


Fig 6.2: VAPT Reporting Portal

CHAPTER 7

REFLECTION AND ANALYSIS

Working on the Next Life: AI Health App project was a valuable and transformative experience. It gave me hands-on exposure to full-stack development using React Native, TypeScript, Node.js, Express.js, and MongoDB, while also introducing me to AI integration using open-source models like BERT. Contributing to both frontend and backend helped me understand how seamless design and strong backend logic work together to deliver a smooth and meaningful user experience in healthcare applications.

This project helped me grow not just technically, but also in terms of collaborative development. I learned how to manage source control with Git and GitHub, participate in peer reviews, and work effectively in a team setting. I also understood the importance of clear communication, consistent code practices, and maintaining documentation to ensure efficient development cycles.

One of the key challenges we faced was ensuring responsiveness across devices and integrating ML models in a way that made predictions fast and accessible. Overcoming these taught me the importance of testing, optimization, and keeping the user at the center of every decision. We aimed to reduce unnecessary hospital visits and offer early, AI-powered health insights that were more accessible and accurate—this core goal made the work feel impactful and purposeful.

Overall, this experience strengthened my full-stack development skills and deepened my understanding of building tech-driven healthcare solutions. It helped me realize how thoughtful design and AI can work together to improve lives, making this internship a truly rewarding learning journey.

CHAPTER 8

RECOMMENDATIONS

1. Enhance AI Model Integration for Broader Use Cases

While the current implementation of AI models like BERT has improved the accuracy of health predictions, expanding the model base to include image-based diagnosis (e.g., CNNs for scans or X-rays) and multilingual NLP support could make the app more inclusive and powerful for a wider audience.

2. Introduce Offline Capabilities

Integrating local storage and offline access for critical health information can be especially helpful for users in remote or low-connectivity areas. Caching essential features and user history locally can improve reliability and accessibility.

3. Implement Role-Based Access Control (RBAC)

As the app handles sensitive medical data, introducing RBAC will allow different levels of access for users, doctors, and administrators. This will enhance data security and ensure that information is appropriately segmented and protected..

4. Continuous Usability Testing

Regular user feedback sessions and usability testing across different devices (phones, tablets) can ensure the app remains responsive, intuitive, and adaptable to real-world use. This will help refine both UI/UX and technical performance.

5. Integration with Wearable Devices and Health APIs

To promote proactive health monitoring, consider integrating APIs from fitness trackers or devices like smartwatches. Real-time data from these sources can enhance predictions and make the app more dynamic.

CHAPTER 9

CONCLUSION

The Next Life: AI Health App project represents a meaningful step forward in bridging the gap between technology and healthcare. By combining mobile and web-based solutions with advanced AI models, we aimed to simplify the way users interact with health services—making diagnostics, consultations, and preventive care more accessible and intelligent. The app is built with a user-first mindset, ensuring not just functionality but also an intuitive, responsive experience that adapts across devices.

Throughout the project, we leveraged a modern tech stack including React Native, TypeScript, Node.js, MongoDB, and cloud services to create a scalable, secure, and real-time system. The integration of machine learning models enabled early health risk predictions by analyzing user reports, potentially providing more accurate and immediate insights than traditional consultations. This aligns with our core vision: reducing unnecessary hospital visits and empowering users to manage their health more proactively.

Collaborating on both frontend and backend development, as well as researching open-source AI models, gave us a holistic understanding of real-world application building. We learned how to balance design, performance, and privacy—especially critical in healthcare environments. The challenges we faced, particularly with cross-platform responsiveness and ML integration, led to innovative solutions and practical learning.

Overall, the project not only enhanced our technical and collaborative abilities but also deepened our sense of purpose. By designing a product that could positively impact lives, we were able to transform academic knowledge into something tangible and socially relevant. Next Life sets the foundation for future advancements in digital health, with the potential to evolve into a highly personalized and AI-powered wellness platform.

CHAPTER 10

REFERENCES

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